No Poker Face — Pre-registration Addendum 2

Confirmatory follow-up: do difficulty requests increase clue leakage, and does a distractor-matching instruction reduce it?

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Parent registration: osf.io/uya5q (Part B, 7 Oct 2026) and Addendum 1 (Gemini writer arm). **Frozen code:** github.com/vjgits/Research-Papers, commit **af87a2757d0c51d8caced9848f098649c02b9176**, folder `No-Poker-Face-Why-AI-Cant-Keep-the-Answer-to-Itself/code-followup/no-poker-face`. **Status:** no follow-up session has been run. The code was tested only offline (mock outputs, never data).

1. Why this follow-up

In the registered main run (168 sessions), both confirmatory hypotheses were not supported. A *secondary, descriptive* result was that questions requested as harder were more answerable without any study material: in invented subjects, knowledge-free accuracy rose from 50.7% (easy) to 58.9% (trickier), while learner-panel accuracy barely moved. Because that pattern was found in the data rather than predicted, we test it here as a confirmatory hypothesis on **fresh data only**, and add a test of a simple remedy. Results will be reported whatever they are.

2. Design

- **Subjects (new, never used before):** Ostrel, an invented card game's tournament rules (21 rules), and the Velloran Glasswork Guild grading code, an invented guild (21 rules). The simulated learner lacks 7 rules in each.
- **Session:** identical to the main run's no-feedback (C) sessions: 7 rounds of 10 four-option questions, round 1 "medium", rounds 2–7 in order A (easy, harder, hard, trickier, medium, harder) or B (the reverse), then a readiness turn. Learner and 3-member panel exactly as in the main run.
- **Arms:**
  - **C (plain):** as the main run.
  - **CFIX:** the same, plus this sentence in every request: *"Make every wrong option as long, as specific and as plausible as the correct one, so the correct answer cannot be spotted from its length, detail or wording."*
- **Confirmatory writers:** claude-opus-5-5 and gpt-5.6-sol (provider defaults). C: 18 sessions per writer × subject (72). CFIX: 10 per writer × subject (40). Orders balanced within every writer × subject × arm cell.
- **Measures per round:** the two cross-family strong solvers (as in the main run's probes) answer the round's questions (a) with no material, which gives **KF**, the knowledge-free accuracy (chance 25%), and (b) with the material, which gives the key check. Panel accuracy (PD), stem length and keyed-option-longest are also recorded. KF of a round is the mean over the two solvers.
- **Exclusion:** a session with fewer than 5 valid questions in any round is dropped (implemented in the code this time).

3. Hypotheses and analysis (confirmatory writers only)

H1 — difficulty requests increase leakage. For each C session, d = mean KF over its rounds requested as hard, trickier or harder, minus KF of its easy round. Test: one-sided one-sample t-test, mean d > 0. Smallest effect of interest: +6 points.

H2 — the instruction reduces leakage. Session KF = mean KF over rounds 2–7. OLS: KF ~ FIX + subject + writer + order, HC3 errors. Test: one-sided, FIX coefficient < 0. Smallest effect of interest: −8 points.

Multiple testing: Holm across H1 and H2. The smaller p is compared with 0.025 and the larger with 0.05 (one-sided).

Power (from the main run's measured variability). H1: session SD of d ≈ 16 points, so n = 72 gives 0.88 power at +6 points. H2: session SD of KF ≈ 11 points, so 40 vs 72 sessions gives 0.95 power at −8 points.

Secondary (descriptive, 95% session bootstrap):

- H1 by writer and by solver.
- d in CFIX, and CFIX minus C.
- KF, PD, stem length and keyed-longest share by level and arm.
- Key check by arm, to see whether the instruction breaks answer keys.

Interpretation rules, fixed now:

- If H1 is rejected, the main run's secondary finding is confirmed on new subjects. If not, we report that it did not replicate.

- If H2 is rejected but the key check in CFIX falls more than 3 points below C, we report that the remedy trades key validity for less leakage.

4. Budget and execution

- **Cost:** estimated from the main run’s measured per-call costs for C sessions: expected **$69**, high case **$90**.
- **Hard cap: $90**. The run proceeds in waves (one replicate of every cell that still needs one), and a wave starts only if its high-case cost fits under the cap, so a budget stop leaves a balanced design.
- **Gemini limits:** about 1,600 calls on each Gemini Flash model (panel and probes), within one day’s limits. No Gemini model writes questions in this follow-up.
- **Deviations:** any deviation will be logged in findings-paper1.md and reported.